

Introduction to Neural Networks

1. Definition and Basic Intuition

- **Deep Learning:** Refers to the process of training Neural Networks (often very large ones).
- **The Single Neuron (Simplest NN):**
 - Can be visualized using a **Linear Regression** model.
 - **Example:** Predicting housing prices (y) based on size (x).
 - Instead of a standard straight line (which might predict negative prices), the function is "bent" to ensure the output is never less than zero.
 - **Mechanism:** Input (x) $\rightarrow$ Neuron (Computation) $\rightarrow$ Output (y).

2. The ReLU Function

- **Name:** Rectified Linear Unit (**ReLU**).
- **Function:** $f(x) = \max(0, x)$.
- **Characteristics:**
 - Output is 0 when the input is negative.
 - Output increases linearly when the input is positive.
 - This is a fundamental building block (activation function) in neural networks.

3. Building Larger Networks

- **Stacking Neurons:** A larger neural network is formed by stacking multiple single neurons together (analogous to stacking Lego bricks).
- **Complex Inputs:** Networks can handle multiple input features simultaneously.
 - *Example Inputs:* Size, Number of Bedrooms, Zip Code, Wealth.
- **Hidden Layers & Feature Extraction:**
 - Intermediate nodes (circles between input and output) are called **Hidden Units**.
 - While humans might conceptually map these units to features like "Family Size" or "Walkability," the network self-determines what each node represents.

4. Architecture and Connectivity

- **Dense Connections:**
 - In a standard neural network layer, the input layer is **densely connected** to the hidden layer.
 - **Rule:** Every input feature connects to every hidden unit.
 - The network is given the input (x) and the target output (y) and learns the intermediate mappings/features automatically.

5. Application Context

- **Supervised Learning:**
 - Neural networks are currently most powerful in **Supervised Learning** contexts.
 - **Goal:** Accurately map Input (x) to Output (y) using labeled training data.
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Supervised Learning with Neural Networks

1. Overview

- **Economic Impact:** The vast majority of economic value currently created by Neural Networks comes from **Supervised Learning**.
- **Core Function:** The goal is to learn a function that maps an Input (x) to an Output (y).

2. Applications of Supervised Learning

Neural networks are effectively applied across various industries by cleverly selecting x and y .

Application	Input ()	Output ()	Note
Real Estate	Home features	Price	Standard prediction
Online Advertising	Ad info, User info	Click (0/1)	Highly lucrative
Photo Tagging	Image	Index (1-1000)	Computer Vision

Application	Input (x)	Output (y)	Note
Speech Recognition	Audio clip	Text transcript	Audio processing
Machine Translation	English sentence	Chinese sentence	NLP
Autonomous Driving	Image, Radar info	Position of other cars	Hybrid system

3. Neural Network Architectures

Different data types require specific Neural Network (NN) architectures for optimal performance.

- **Standard Neural Networks:**
 - Used for general tabular data.
 - *Examples:* Real Estate, Online Advertising.
- **Convolutional Neural Networks (CNN):**
 - Specialized for **Image Data**.

1 | * *Examples:* Photo tagging, Object detection.

- **Recurrent Neural Networks (RNN):**
 - Specialized for **Sequence Data** (data with a temporal component).

1 | * *Examples:* Audio (1D time series), Language/Text (sequence of words).

- **Hybrid Architectures:**
 - Used for complex tasks combining different data types.
 - *Example:* Autonomous driving (Images + Radar).

4. Structured vs. Unstructured Data

Machine Learning is applied to two distinct categories of data.

- **Structured Data:**
 - **Definition:** Data found in databases with well-defined meanings for each feature (column).

- *Examples:* Housing prices database (Size, #Bedrooms), User data (Age, Ad ID).
 - *Economic Value:* Creates significant short-term economic value (e.g., better ad systems, recommendations).
 - **Unstructured Data:**
 - **Definition:** Raw data types where features are not explicitly defined as columns.
 - *Examples:* Audio (raw wave), Images (pixel values), Text (individual words).
 - *Human vs. Machine:* Humans are naturally empathetic and good at understanding unstructured data. Historically, computers struggled with this.
 - *Impact of Deep Learning:* Neural Networks have drastically improved computer performance on unstructured data (Speech recognition, Image recognition), enabling new applications.
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Why is Deep Learning Taking Off?

1. The Scale Driver: Data

The primary driver for the recent surge in Deep Learning performance is the massive increase in available data.

- **Performance vs. Data Graph:**
 - **Traditional Algorithms (e.g., SVM, Logistic Regression):** Performance improves with data up to a point but eventually **plateaus**. They cannot effectively utilize huge amounts of data.
 - **Neural Networks:** Performance continues to increase as the amount of data increases.
 - **Network Size:**
 - **Small NN:** Performance is better than traditional algorithms but may still plateau early.
 - **Medium NN:** Better performance.
 - **Large NN:** Dominates performance on very large datasets; rarely plateaus.
- **Sources of Data:**
 - Digitization of society (web activity, mobile apps).

- Inexpensive sensors (cameras, accelerometers, IoT).
- **Notation:**
 - m : The size of the training set (number of labeled training examples).
- **Data Regimes:**
 - **Small Data Regime (Low m):** The ordering of algorithms is undefined. Performance depends heavily on **feature engineering** skill rather than the algorithm itself (an SVM might beat a Neural Network here).
 - **Big Data Regime (High m):** Large Neural Networks consistently dominate other approaches.

2. The Computation Driver

- **Hardware:** The ability to train very large neural networks has been enabled by faster **CPUs** and specialized hardware like **GPUs**.
- **Impact:** Computation power allows us to scale up the size of the neural networks (number of hidden units, parameters, and connections).

3. The Algorithmic Driver

Algorithmic innovation has largely focused on making neural networks run **faster**, which indirectly enables the use of more data and larger networks.

- **Key Example: Activation Functions**
 - **Sigmoid Function:** Historically used but problematic. It has regions where the slope (gradient) is nearly zero, causing learning to be very slow (the Vanishing Gradient problem).
 - **ReLU (Rectified Linear Unit):**
 - Gradient is 1 for all positive input values.
 - Gradient is much less likely to shrink to zero.
 - **Result:** Switching from Sigmoid to ReLU makes **Gradient Descent** work much faster.

4. The Iterative Cycle of Research

Fast computation is crucial not just for the final result, but for the **research workflow**.

- **The Cycle:** Idea → Code → Experiment → Result.
- **Productivity:**
 - Slow computation (e.g., waiting a month for a result) hinders progress.
 - Fast computation (e.g., results in 10 minutes or a day) allows practitioners

to iterate quickly, test more ideas, and discover effective architectures faster.

Summary of Drivers

The rise of Deep Learning is fueled by a positive feedback loop of:

1. **Data:** Increasing digitization and sensor availability.
 2. **Computation:** Specialized hardware (GPUs).
 3. **Algorithms:** Innovations that optimize speed (e.g., ReLU).
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